



Accenture Computer Networking — Tier 2

Tier 2 focuses on the networking concepts that are very useful for Accenture MCQs after OSI, TCP/UDP, IP, devices, and common protocols.

1. MAC Address vs IP Address ★★★★★

This is one of the most common areas of confusion.

MAC Address

MAC = Media Access Control address

A MAC address is a hardware/link-layer address associated with a network interface.

Example: 00:1A:2B:3C:4D:5E

Important facts:

- Primarily used at Data Link Layer (Layer 2)
- Used for communication on the local network
- Ethernet frames contain source and destination MAC addresses
- Typically represented in hexadecimal

IP Address

An IP address is a logical address used for communication across IP networks.

Example: 192.168.1.10

Important facts:

- Network layer
- Used for routing
- Can change depending on network
- IPv4 = 32 bits
- IPv6 = 128 bits

🔥 MAC vs IP

Feature	MAC	IP
Layer	L2 (Data Link)	L3 (Network)
Type	Link-layer address	Logical/network address
Main use	Local network delivery	Routing between networks
Example	00:1A:2B:3C:4D:5E	192.168.1.10
Format	Hexadecimal	IPv4 decimal / IPv6 hexadecimal
Typically changes?	Usually stable for interface	Can change between networks

Easy memory:

- **MAC** = Local delivery
- **IP** = Network routing

2. ARP ★★★★★

Definition

ARP = Address Resolution Protocol

ARP is used in IPv4 networks to determine the MAC address associated with a known IPv4 address on the local network.

Suppose PC A knows IP = 192.168.1.20, but doesn't know MAC = ?. It uses ARP.

ARP Process

Step 1: PC A broadcasts: "Who has 192.168.1.20?"

```
PC A
  ↓
ARP Request (Broadcast)
  ↓↓↓↓
LAN
  ↓
192.168.1.20
```

Step 2: The device with that IP replies: "192.168.1.20 is at AA:BB:CC:DD:EE:FF." (ARP Reply - Unicast)

PC A stores the mapping in its: **ARP cache/table**

Important ARP fact:

ARP maps **IPv4 address** → **MAC address**.

Don't say MAC → IP for the normal ARP operation (that is RARP/BOOTP).

Accenture MCQ Highlight:

*"A computer knows the destination's IPv4 address but needs its MAC address to send an Ethernet frame on the local network. Which protocol is used?" → **ARP***

3. ARP vs DNS

Easy confusion. Know the distinction clearly:

```
ARP:
IPv4 address → MAC address

DNS:
Domain name → IP address

So:
ARP → IP → MAC
DNS → Name → IP
```

Example Workflow:

```
google.com
  ↓ DNS
142.x.x.x
  ↓ ARP (if destination is local/on-link)
MAC address
```

4. ICMP ★★★★★

ICMP = Internet Control Message Protocol

ICMP is used for:

- Error reporting
- Diagnostics
- Network control/information

It is associated with the IP/Network layer.

Ping

The most famous use of ICMP is **ping**.

When you run ping 8.8.8.8, the system commonly uses:

```
ICMP Echo Request
      ↓
    Server
      ↓
ICMP Echo Reply
```

If the reply arrives: The destination is reachable at the network level, subject to the target/network allowing ICMP.

Important: ICMP is not TCP or UDP. It doesn't use a TCP/UDP port in the normal sense.

Accenture trap: "Which protocol is commonly used by ping?" → **ICMP**

5. Ports ★★★★★

A port number identifies a network service/application endpoint on a host.

IP identifies the host/interface. Port helps identify: *Which service/process should receive the traffic?*

Example: 192.168.1.10 : 443 (Device at 192.168.1.10, HTTPS service at port 443).

IP vs Port (Apartment Analogy):

- **IP address** = Building address (Which machine?)
- **Port** = Apartment/service number (Which service?)

Common Ports Table

Service	Port	Transport	Service	Port	Transport
FTP	21	TCP	HTTP	80	TCP
SSH	22	TCP	POP3	110	TCP
Telnet	23	TCP	IMAP	143	TCP
SMTP	25	TCP	HTTPS	443	TCP
DNS	53	UDP/TCP	DHCP	67/68	UDP

6. Socket ★★☆☆

A socket is an endpoint for network communication.

For basic MCQs, think: **IP address + Port = Endpoint**

Example: 192.168.1.10:443 (Identifies an endpoint associated with HTTPS communication).

A TCP connection is commonly identified by a combination involving:

- Source IP
- Source Port
- Destination IP
- Destination Port
- Protocol

For Accenture, understand the basic concept rather than going into kernel-level socket implementation.

7. LAN / MAN / WAN / PAN ★★☆☆

Network types are classified partly by geographic coverage.

- **PAN (Personal Area Network):** Very small area around a person (~a few meters). Example: Phone ↔ Bluetooth ↔ Earbuds.
- **LAN (Local Area Network):** Covers a small geographic area such as Home, Office, Building, Lab. Example: PCs connected to a local Switch.
- **MAN (Metropolitan Area Network):** Covers a larger area such as a city/metropolitan region connecting multiple buildings.
- **WAN (Wide Area Network):** Covers very large geographical areas (Country, Multiple countries, Global network). The **Internet** is the most prominent example of a global WAN/internet network.

🔥 PAN vs LAN vs MAN vs WAN Summary:

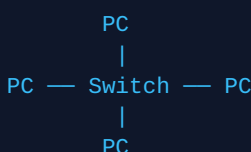
- **PAN** → Person (~meters)
- **LAN** → Building/office
- **MAN** → City
- **WAN** → Country/world

8. Network Topologies ★★☆☆

Topology describes how network devices are arranged/connected. Important types: **Bus, Star, Ring, Mesh, Tree**.

Star Topology ★★☆☆

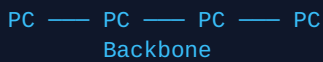
All devices connect to a central device, typically a switch in modern Ethernet LANs.



- **Advantage:** If one cable/device fails, other devices can generally continue operating.
- **Disadvantage:** If the central device fails, large part of the network can be affected.
- **Status:** Most common modern LAN topology.

Bus Topology

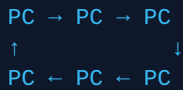
All devices share a common backbone.



- **Problem:** A backbone failure can affect the entire network. Older Ethernet networks commonly used bus-like arrangements.

Ring Topology

Devices form a closed ring. Data travels around the ring according to the technology.



Mesh Topology

Devices have multiple interconnections. In a **Full Mesh**, every device connects directly to every other device.

For n devices, number of links = $n(n - 1) / 2$.

Example: For 4 devices: $4(4 - 1) / 2 = 6$ links.

- **Advantage:** Very high redundancy and fault tolerance.
- **Disadvantage:** Expensive and complex cabling.

9. Subnetting ★★★★★

This is the most technical Tier-2 topic. You should know the basic concepts and simple calculations for Accenture.

What is Subnetting?

Subnetting divides one IP network into smaller logical networks called **Subnets**.

Example: 192.168.1.0/24 can be divided into smaller networks.

Why? Better IP utilization, Network organization, Traffic isolation, Smaller broadcast domains.

Subnet Mask & CIDR

A subnet mask determines which portion of an IP address identifies the network and which portion identifies hosts.

- IP: 192.168.1.10 | Mask: 255.255.255.0
- Equivalent CIDR notation: 192.168.1.10/24
- **CIDR** = Classless Inter-Domain Routing. The /24 is the **Prefix length** (First 24 bits are network prefix, remaining $32 - 24 = 8$ bits are host bits).

Host Calculation & Formulas

- **Host bits:** $h = 32 - \text{prefix}$
- **Total addresses:** 2^h
- **Traditional usable hosts:** $2^h - 2$ (Subtract Network address and Broadcast address)

Example: 192.168.1.0/24

- Host bits = $32 - 24 = 8$
- Total addresses = $2^8 = 256$
- **Network address:** 192.168.1.0
- **First usable host:** 192.168.1.1
- **Last usable host:** 192.168.1.254

- **Broadcast address:** 192.168.1.255
- **Usable hosts:** $256 - 2 = 254$

Common CIDR Values Table

CIDR	Subnet Mask	Host bits	Total addresses	Traditional usable hosts
/24	255.255.255.0	8	256	254
/25	255.255.255.128	7	128	126
/26	255.255.255.192	6	64	62
/27	255.255.255.224	5	32	30
/28	255.255.255.240	4	16	14
/29	255.255.255.248	3	8	6
/30	255.255.255.252	2	4	2 (Point-to-point)

10. Default Gateway ★★★★★

Definition: A default gateway is the router/interface a host uses to reach destinations outside its local subnet when no more specific route is available.

- If communicating with 192.168.1.20 (same subnet) → direct communication at local network level.
- If communicating with 8.8.8.8 (outside subnet) → traffic sent to **Default Gateway**.

Easy memory: Outside my network → Default gateway

11. NAT — Deeper Understanding ★★★★★

Scenario where multiple private PCs connect to the internet:

```
PC1 (192.168.1.10)  |
PC2 (192.168.1.11)  |——→ NAT/PAT Router ——→ Public IP ——→ Internet
PC3 (192.168.1.12)  |
```

PAT (Port Address Translation / NAT Overload): Allows multiple private hosts to share one public IPv4 address by using different source port mappings.

- **NAT:** Translates IP addresses.
- **PAT:** Uses port numbers to allow many private addresses to share one public address.

Accenture level: NAT = Address translation | PAT = Address + Port translation (many-to-one sharing).

12. Unicast / Broadcast / Multicast ★★★★★

These describe how packets are delivered across the network:

- **Unicast:** One sender → One receiver (1 → 1). *Example:* Requesting a webpage from a server.
- **Broadcast:** One sender → All devices in the relevant broadcast domain (1 → ALL). *Example:* 255 . 255 . 255 . 255 in IPv4. Routers normally do not forward ordinary IPv4 broadcasts between networks.
- **Multicast:** One sender → Multiple interested receivers (1 → GROUP). Only devices that explicitly joined the multicast group receive the traffic.

Type	Sender	Receivers
Unicast	1	1 (Specific host)
Broadcast	1	All in broadcast domain
Multicast	1	Selected group

Easy memory: Uni → One | Broad → Everyone | Multi → Group

Accenture Computer Networking — Tier 2 MCQ Practice Bank

Q1. MAC Address

A computer needs to communicate with another device on the same Ethernet LAN. Which address is primarily used for frame delivery at Layer 2?

- A. Port number
- B. MAC address
- C. Domain name
- D. URL

Answer: B. MAC address ✓

A device communicating on the same Ethernet LAN needs the destination's MAC address for Layer-2 frame delivery.

- A. *Port number* ✗ — Identifies a service/application endpoint.
- B. *MAC address* ✓ — Layer-2 address used for local-link delivery.
- C. *Domain name* ✗ — Human-readable name resolved by DNS.
- D. *URL* ✗ — Identifies a web resource.

 Local Ethernet delivery → MAC

Q2. IP Address

Which address is primarily used for logical addressing and routing packets between different networks?

- A. MAC address
- B. IP address
- C. Port number
- D. DNS name

Answer: B. IP address ✓

- A. *MAC address* ✗ — Primarily used for local-link communication.
- B. *IP address* ✓ — Logical addressing and routing between networks.
- C. *Port number* ✗ — Identifies a service.
- D. *DNS name* ✗ — Domain names are resolved to IP addresses.

 **Routing → IP**

Q3. MAC vs IP

Which statement is correct?

- A. MAC addresses are used for routing across the Internet, while IP addresses are only used inside a LAN
- B. IP addresses are Layer-2 addresses, while MAC addresses are Layer-3 addresses
- C. MAC addresses are primarily used for local-link delivery, while IP addresses are used for logical addressing and routing
- D. MAC and IP addresses are exactly the same thing

Answer: C. MAC addresses are primarily used for local-link delivery, while IP addresses are used for logical addressing and routing ✓

- A. ✗ — MAC addresses aren't used for Internet-wide routing.
- B. ✗ — Layers are reversed.
- C. ✓ *Correct* — MAC is Layer 2, IP is Layer 3.
- D. ✗ — MAC and IP addresses serve different purposes.

Remember: MAC → Layer 2 → Local delivery | IP → Layer 3 → Routing

Q4. ARP

A computer knows that the destination is 192 . 168 . 1 . 20, but it doesn't know the destination's MAC address. What should it use?

- A. DNS
- B. DHCP
- C. ARP
- D. ICMP

Answer: C. ARP ✓

The computer knows IP = 192 . 168 . 1 . 20, but MAC = ?. ARP finds the MAC address corresponding to the IPv4 address on the local network.

- A. *DNS* ✗ — Domain name → IP.
- B. *DHCP* ✗ — Provides network configuration.
- C. *ARP* ✓ — IPv4 → MAC.
- D. *ICMP* ✗ — Diagnostics/error reporting.

 **ARP = Address Resolution Protocol**

Q5. ARP Request

What type of transmission is normally used by an IPv4 ARP request on an Ethernet LAN?

- A. Unicast
- B. Broadcast
- C. Multicast
- D. Anycast

Answer: B. Broadcast ✓

An ARP request is normally sent as an Ethernet broadcast on the local LAN ("Who has 192.168.1.20?").

- A. *Unicast* ✗ — One specific receiver.
- B. *Broadcast* ✓ — All devices in the local broadcast domain receive the request.
- C. *Multicast* ✗ — Selected group.
- D. *Anycast* ✗ — One of multiple possible destinations, not normal ARP request mechanism.

Q6. ARP Reply

After receiving an ARP request, the device that owns the requested IPv4 address typically sends the ARP reply as:

- A. Unicast to the requester
- B. Broadcast to all networks
- C. Multicast to the Internet
- D. Anycast

Answer: A. Unicast to the requester ✓

After the device discovers "192.168.1.20 is my IP", it generally sends the ARP reply directly to the requesting device.

- A. *Unicast* ✓
- B. *Broadcast* ✗ — Normal ARP reply is not broadcast.
- C. *Multicast* ✗
- D. *Anycast* ✗

Easy memory: ARP Request → Broadcast | ARP Reply → Usually Unicast

Q7. DNS vs ARP

Which mapping is correct?

- A. DNS: IP → MAC
- B. ARP: Domain → IP
- C. DNS: Domain → IP
- D. ARP: Domain → IP

Answer: C. DNS: Domain → IP ✓

- A. *DNS: IP → MAC* ✗ — That's not DNS.
- B. *ARP: Domain → IP* ✗
- C. *DNS: Domain → IP* ✓
- D. *ARP: Domain → IP* ✗

Memorize: DNS → google.com → IP | ARP → IP → MAC

Q8. ICMP

Which protocol is commonly used by the ping command?

- A. TCP
- B. UDP
- C. ICMP
- D. ARP

Answer: C. ICMP ✓

The ping command commonly uses: **ICMP Echo Request / Echo Reply**.

- A. TCP ✗
- B. UDP ✗
- C. ICMP ✓
- D. ARP ✗

 **Ping → ICMP**

Q9. ICMP Purpose

What is a major purpose of ICMP?

- A. File transfer
- B. Email delivery
- C. Network diagnostics and error reporting
- D. Assigning IP addresses

Answer: C. Network diagnostics and error reporting ✓

- A. File transfer ✗ → FTP.
- B. Email delivery ✗ → SMTP.
- C. Network diagnostics/error reporting ✓ → ICMP.
- D. Assigning IP addresses ✗ → DHCP.

Examples of ICMP use: Ping, Destination unreachable, Time exceeded.

Q10. Port Number

A server has the address: 192.168.1.10:443. What does 443 represent?

- A. MAC address
- B. Network address
- C. Port number
- D. Subnet mask

Answer: C. Port number ✓

Given: 192.168.1.10:443, 443 is the port.

- A. MAC address ✗
- B. Network address ✗
- C. Port number ✓
- D. Subnet mask ✗

 **IP : Port**

Q11. IP vs Port

Which statement is most accurate?

- A. IP identifies a host/interface, while a port identifies a service/application endpoint
- B. IP identifies the application, while port identifies the physical cable
- C. IP and port are both MAC addresses
- D. Port identifies the network, while IP identifies the protocol

Answer: A. IP identifies a host/interface, while a port identifies a service/application endpoint ✓

Example: 192.168.1.10:443 (IP = Host, Port = Service).

- A. ✓ Correct
- B. ✗ Port doesn't identify physical cables.
- C. ✗ Neither is necessarily a MAC address.
- D. ✗ Reversed/wrong.

🧠 IP = Which host? Port = Which service?

Q12. LAN

A network connects computers within a university laboratory building. What type of network is this?

- A. WAN
- B. MAN
- C. LAN
- D. PAN

Answer: C. LAN ✓

A university laboratory within one building is a Local Area Network.

- A. WAN ✗ — Large geographic area.
- B. MAN ✗ — Metropolitan/city scale.
- C. LAN ✓
- D. PAN ✗ — Personal area.

Q13. WAN

A company connects offices located in India, the USA, and Germany. What type of network is most appropriate?

- A. PAN
- B. LAN
- C. MAN
- D. WAN

Answer: D. WAN ✓

Offices in India, USA, and Germany require communication across huge geographic distances.

- A. PAN ✗
- B. LAN ✗
- C. MAN ✗
- D. WAN ✓

🧠 Large geographic area → WAN

Q14. MAN

A network connects several offices spread across the same metropolitan city. Which type of network is this?

- A. PAN
- B. LAN
- C. MAN
- D. WAN

Answer: C. MAN ✓

Several offices within the same metropolitan city.

- A. *PAN* ✗ — Personal devices.
- B. *LAN* ✗ — Smaller local area.
- C. *MAN* ✓ — Metropolitan/city area.
- D. *WAN* ✗ — Larger geographic coverage.

Q15. PAN

A smartphone communicates with wireless earbuds using Bluetooth. This is an example of:

- A. PAN
- B. LAN
- C. MAN
- D. WAN

Answer: A. PAN ✓

Phone + Bluetooth earbuds is a personal-area network.

- A. *PAN* ✓
- B. *LAN* ✗
- C. *MAN* ✗
- D. *WAN* ✗

 **Personal devices → PAN**

Q16. Star Topology

In a network, all computers connect to a central switch. Which topology is this?

- A. Bus
- B. Ring
- C. Star
- D. Mesh

Answer: C. Star ✓

All devices connect to a central switch.

- A. *Bus* ✗ — Shared backbone.
- B. *Ring* ✗ — Circular connection.
- C. *Star* ✓
- D. *Mesh* ✗ — Multiple interconnections.

Q17. Star Failure

In a star topology, what happens if the central switch fails?

- A. Only one connected computer is affected
- B. The entire connected portion of the network can be disrupted
- C. Nothing happens
- D. The network automatically becomes a bus topology

Answer: B. The entire connected portion of the network can be disrupted ✓

The central switch is a critical single point of failure.

- A. ✗ — Not necessarily only one computer.
- B. ✓ *Correct*
- C. ✗ — The network can lose connectivity.
- D. ✗ — It doesn't automatically become a bus.

🧠 **Star's major weakness** → **Central device failure**

Q18. Mesh Topology

Which topology provides high redundancy because devices have multiple interconnections?

- A. Bus
- B. Star
- C. Ring
- D. Mesh

Answer: D. Mesh ✓

Mesh provides multiple paths between devices.

- A. *Bus* ✗ — Shared backbone.
- B. *Star* ✗ — Central device.
- C. *Ring* ✗ — Circular path.
- D. *Mesh* ✓ — Multiple interconnections/redundancy.

🧠 **High redundancy** → **Mesh**

Q19. Subnetting

What is the primary purpose of subnetting?

- A. Convert domain names into IP addresses
- B. Divide a network into smaller logical networks
- C. Encrypt network traffic
- D. Replace MAC addresses

Answer: B. Divide a network into smaller logical networks ✓

- A. ✗ DNS's job.
- B. ✓ Subnetting divides networks into smaller subnets.
- C. ✗ Encryption protects data.
- D. ✗ Doesn't replace MAC addresses.

Benefits: Better IP utilization, Organization, Smaller broadcast domains, Network segmentation.

Q20. CIDR /24

What does /24 mean in: 192.168.1.0/24?

- A. 24 host bits
- B. 24 network-prefix bits
- C. 24 total bits
- D. 24 available hosts

Answer: B. 24 network-prefix bits ✓

192.168.1.0/24 → /24 is the Prefix length.

- A. 24 host bits ✗
- B. 24 network-prefix bits ✓
- C. 24 total bits ✗ — IPv4 has 32 total bits.
- D. 24 available hosts ✗

Therefore: 32 total bits – 24 network bits = 8 host bits.

Q21. /24 Total Addresses

How many total IPv4 addresses are in a typical /24 subnet?

- A. 64
- B. 128
- C. 256
- D. 512

Answer: C. 256 ✓

Host bits: $32 - 24 = 8$

Total: $2^8 = 256$

- A. 64 ✗
- B. 128 ✗
- C. 256 ✓
- D. 512 ✗

Q22. /24 Usable Hosts

Traditionally, how many usable host addresses are available in a /24 IPv4 subnet?

- A. 252
- B. 254
- C. 256
- D. 128

Answer: B. 254 ✓

Traditional calculation:

$2^8 = 256$ addresses

$256 - 2 = 254$ usable hosts

Why subtract 2? 1 for Network address, 1 for Broadcast address.

- A. 252 ✗
- B. 254 ✓
- C. 256 ✗ — Total addresses, not traditional usable hosts.
- D. 128 ✗

Q23. /26 Total Addresses

How many total IPv4 addresses are in a /26 subnet?

- A. 32
- B. 64
- C. 128
- D. 256

Answer: B. 64 ✓

Host bits = $32 - 26 = 6$

Total = $2^6 = 64$

- A. 32 ✗
- B. 64 ✓
- C. 128 ✗
- D. 256 ✗

Q24. /26 Usable Hosts

Traditionally, how many usable host addresses are available in a /26 subnet?

- A. 62
- B. 64
- C. 30
- D. 126

Answer: A. 62 ✓

Total = $2^6 = 64$

Usable = $64 - 2 = 62$

- A. 62 ✓
- B. 64 ✗ — Total addresses.
- C. 30 ✗ — Traditional usable hosts for /27.
- D. 126 ✗ — Traditional usable hosts for /25.

Q25. Network Address

For the subnet: 192.168.1.0/24, which is the network address?

- A. 192.168.1.0
- B. 192.168.1.1
- C. 192.168.1.254
- D. 192.168.1.255

Answer: A. 192.168.1.0 ✓

- A. 192.168.1.0 ✓ — Network address.
- B. 192.168.1.1 ✗ — First traditional usable host.
- C. 192.168.1.254 ✗ — Last traditional usable host.
- D. 192.168.1.255 ✗ — Broadcast address.

Full picture: 192.168.1.0 (Network) → 192.168.1.1 (First host) ... 192.168.1.254 (Last host) → 192.168.1.255 (Broadcast).

Q26. Broadcast Address

For: 192.168.1.0/24, which is the broadcast address?

- A. 192.168.1.0
- B. 192.168.1.1
- C. 192.168.1.254
- D. 192.168.1.255

Answer: D. 192.168.1.255 ✓

For 192.168.1.0/24:

- A. .0 ✗ → Network address.
- B. .1 ✗ → First traditional usable host.
- C. .254 ✗ → Last traditional usable host.
- D. .255 ✓ → Broadcast address.

🧠 For a /24 ending in .0: .0 = Network, .255 = Broadcast

Q27. Default Gateway

A PC wants to communicate with a server located outside its local subnet. Where will the PC normally send the packet first?

- A. DNS server
- B. Default gateway
- C. DHCP server
- D. Switch's MAC table

Answer: B. Default gateway ✓

If the destination is outside the local subnet, the host sends the traffic toward its default gateway.

- A. *DNS server* ✗ — Resolves names.
- B. *Default gateway* ✓ — Provides next-hop path out of the local network.
- C. *DHCP server* ✗ — Provides network configuration.
- D. *Switch's MAC table* ✗ — Used by switches for Layer-2 forwarding.

🧠 Outside local network → Default gateway

Q28. NAT

A home router allows several devices with private IP addresses to access the Internet using a public IP address. Which technology enables this address translation?

- A. ARP
- B. NAT
- C. ICMP
- D. DNS

Answer: B. NAT ✓

Private addresses are translated to a public address for Internet communication.

- A. *ARP* ✗ — IPv4 → MAC.
- B. *NAT* ✓ — Network Address Translation.
- C. *ICMP* ✗ — Diagnostics/error reporting.
- D. *DNS* ✗ — Name resolution.

Q29. PAT

Twenty private devices access the Internet using the same public IPv4 address, with different port mappings. Which technology is most closely associated with this?

- A. PAT
- B. ARP
- C. ICMP
- D. DHCP

Answer: A. PAT ✓

Multiple private devices share one public IPv4 address using different port mappings.

- A. *PAT* ✓ — Port Address Translation.
- B. *ARP* ✗
- C. *ICMP* ✗
- D. *DHCP* ✗

```
192.168.1.10:5000 }  
192.168.1.11:5001 } → PAT → Public IP  
192.168.1.12:5002 }
```

🧠 Many private devices → One public IP + different ports → PAT

Q30. 🔥 Scenario — Unicast/Broadcast/Multicast

A video server sends the same stream to a specific group of subscribed receivers rather than to every device on the network. What type of communication is most appropriate?

- A. Unicast
- B. Broadcast
- C. Multicast
- D. Anycast

Answer: C. Multicast ✓

The server sends data to a specific group of interested receivers.

- A. *Unicast* ✗ — One sender → one receiver.
- B. *Broadcast* ✗ — One sender → all devices in the relevant broadcast domain.
- C. *Multicast* ✓ — One sender → selected group.
- D. *Anycast* ✗ — One sender sends to the nearest/best member of a group of possible receivers.

Remember: Unicast → 1 → 1 | Broadcast → 1 → ALL | Multicast → 1 → GROUP